Day 117: Imbalanced Data — SMOTE Against Class Weighting

MD Mutasim Billah | 52-Week Data Science to ML/AI Roadmap | Week 17, Day 6 | Complete study guide

Watch: SMOTE Explained with Example (At A Glance!) | Handling data imbalance using SMOTE in python (Statistics and Data science)

Script: `day117_imbalanced_data.py` · every number in this guide is copied by program from `day117_run_output.txt`, the real output of that script.

When one class is rare, a model can score 90% accuracy by never predicting it. Six remedies are written from scratch and compared on three real rare-class problems with two models. The useful result is what they change. On Diabetes, logistic regression finds 0.2750 of the rare cases untreated and about 0.7929 with class weights, while its ranking quality (ROC AUC) barely moves (0.8967 against 0.8991): most of what rebalancing does is move the decision threshold. SMOTE and class weights end up close; the price is inflated probabilities (0.2670 predicted against a true 0.1061).

0. Where today fits

DEFINITION — start here, no prior knowledge assumed

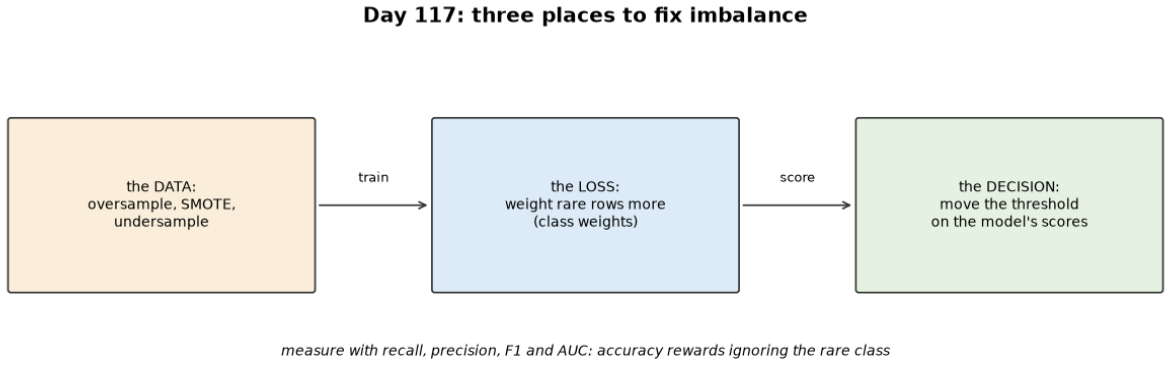
Imbalanced data has one class much rarer than the other. **Recall** is the share of rare rows a model finds; **precision** is the share of its alarms that are right; **F1** balances the two. **ROC AUC** is the chance that a random rare row scores above a random common one: it judges the ranking, at every threshold at once.

SMOTE (synthetic minority oversampling) invents new rare rows on the line between a rare row and one of its rare neighbours. **Class weights** instead make each rare row count more in the loss.

REAL-WORLD SCENARIO

Fraud screening: one transaction in a hundred is fraud. A system that approves everything is 99% accurate and catches nothing; the question is how many frauds it catches and how many good customers it blocks.

SYSTEM DIAGRAM



Three places to act: the data, the loss, or the decision threshold.

Today's part for the week's build

`day117_exercise.py` is the metrics and SMOTE, checked by hand on six rows. Its real output:

ACTUAL OUTPUT

```
perfect ranking: AUC 1.000; reversed: 0.000; all tied: 0.500
rare rows ranked 1st and 3rd: AUC by hand 7 of 8 pairs = 0.875 -> 0.875; average precision by hand
    (1/1 + 2/3) / 2 = 0.833 -> 0.833
at threshold 0.5: precision 0.500 (1 of 2 flagged), recall 0.500 (1 of 2 found), F1 0.500,
    balanced accuracy 0.625
SMOTE, 5 new points from a triangle of rare points: all on its edges or inside: True; first:
    (1.927, 0.000)
```

Data, all real rows: digits (a 9 against the rest, 10% as the data comes), breast cancer with malignant tumours subsampled to 5%, and the top 10% of Diabetes progression scores. Models: Day 116's logistic regression and Day 107's forest. scikit-learn's metrics are an outside check.

How to read each section

Step	What it gives you
DEFINITION	What the idea is, in plain words. Start here.
REAL-WORLD SCENARIO	An everyday situation where the same idea shows up.
MATHEMATICS	The formula, when the section introduces one.
SPECIFICATION	The exact setup of the experiment.
IMPLEMENTATION	The real code from the script.
ACTUAL OUTPUT	What the script really printed, copied unchanged.
What it means	How to read the numbers, and what they do and do not show.
Why choose this? / Why NOT this?	When the idea is the right tool, and what it costs.

1. The accuracy paradox

DEFINITION — start here, no prior knowledge assumed

Always predicting the common class scores the common share as accuracy, and finds none of the rare rows.

REAL-WORLD SCENARIO

A smoke alarm with no battery never gives a false alarm.

ACTUAL OUTPUT

```
problem                       rare share  always common: acc  recall  logistic, as is: acc  recall
Digits: 9 against the rest        0.1002              0.8998  0.0000                0.9926  0.9630
Breast cancer, 5% malignant       0.0494              0.9469  0.0000                0.9912  0.8333
Diabetes: top 10%                 0.1065              0.8939  0.0000                0.8788  0.1429
a model that never predicts the rare class is highly 'accurate' and useless: recall, the share of rare rows
    found, is the number to watch
Actual output from this lesson's run.
```

What it means

'Always common' scores 0.8998, 0.9469 and 0.8939 accuracy with zero recall. Plain logistic regression on Diabetes is even less accurate (0.8788) and finds only 0.1429 of the rare cases on this split. Accuracy is the wrong scoreboard here.

WHY CHOOSE THIS?

Shows at a glance why accuracy is not enough.

WHY NOT THIS? (TRADEOFFS)

One split each; section 4 uses 20.

2. The metrics, checked

DEFINITION — start here, no prior knowledge assumed

ROC AUC computed from ranks (the Mann-Whitney form), average precision from precision at each rare row, F1 from the confusion counts.

REAL-WORLD SCENARIO

Checking a home-made ruler against a shop-bought one.

MATHEMATICS

$$precision = TP/(TP+FP),\ recall = TP/(TP+FN),\ F1 = 2PR/(P+R)$$

$$AUC = (\text{sum of the rare rows' ranks} - n_1(n_1-1)/2) / (n_1\, n_0)$$

IMPLEMENTATION

```
def roc_auc(y, p):
    r = np.argsort(np.argsort(p, kind="stable"), kind="stable").astype(float)
    for v in np.unique(p):
        m = p == v
        if m.sum() > 1:
            r[m] = r[m].mean()
    n1 = y.sum(); n0 = len(y) - n1
    return float((r[y == 1].sum() - n1 * (n1 - 1) / 2) / (n1 * n0))
def avg_precision(y, p):
    o = np.argsort(-p, kind="stable"); hits = y[o] == 1
    prec = np.cumsum(hits) / np.arange(1, len(y) + 1)
    return float(prec[hits].mean())
```

ACTUAL OUTPUT

```
digits, one split: ROC AUC 0.997595 against 0.997595; average precision 0.985937 against 0.985937; F1 0.962963
    against 0.962963
with scores rounded to one decimal (many ties): ROC AUC 0.989290 against 0.989290; average precision 0.977187
    against 0.976102 (this version ignores ties)
Actual output from this lesson's run.
```

What it means

All three match scikit-learn to six decimals (0.997595, 0.985937, 0.962963). With heavy ties, AUC still matches (ties count half), but this simple average precision drifts (0.977187 against 0.976102): it ignores ties, and the script says so.

WHY CHOOSE THIS?

Every number in the lesson rests on checked metrics.

WHY NOT THIS? (TRADEOFFS)

Average precision here is not tie-aware.

3. SMOTE by hand

DEFINITION — start here, no prior knowledge assumed

Four rare points, k = 1: each new point lies somewhere on the segment from a rare point to its nearest rare neighbour.

REAL-WORLD SCENARIO

Filling gaps on a map by drawing between known towns, without checking whether the road crosses a lake.

IMPLEMENTATION

```
def smote(X, k, n_new, rng):
    D = d115.sqdist(X, X); np.fill_diagonal(D, np.inf)
    nn = np.argsort(D, 1)[:, :k]
    i = rng.randint(0, len(X), n_new); j = nn[i, rng.randint(0, min(k, len(X) - 1), n_new)]
    return X[i] + rng.uniform(size=(n_new, 1)) * (X[j] - X[i])
```

ACTUAL OUTPUT

```
rare points (0,0), (1,0), (0,1), (5,5); k = 1, so each new point lies between a rare point and its
    nearest rare neighbour
  new point (0.545, 0.000)
  new point (3.305, 2.882)
  new point (0.354, 0.000)
  new point (0.438, 0.000)
  new point (1.433, 0.541)
  new point (1.145, 0.182)
every new point lies on such a segment: True; (5,5)'s nearest rare neighbour is far away, so its
    new points cross empty space
Actual output from this lesson's run.
```

What it means

Every new point lies on a segment (True). The point (5, 5) shows SMOTE's weakness: its nearest rare neighbour is far away, so a new point such as (3.305, 2.882) lands in empty space where no real rare row has ever been.

WHY CHOOSE THIS?

New rare rows that are not exact copies.

WHY NOT THIS? (TRADEOFFS)

It invents data between points that may not belong together.

4. Six remedies, two models, three problems

DEFINITION — start here, no prior knowledge assumed

Every remedy, with logistic regression and with a forest, on 20 splits. The test rows keep their natural (or stated) rare share.

REAL-WORLD SCENARIO

Six treatments, two patient groups, three hospitals.

SPECIFICATION

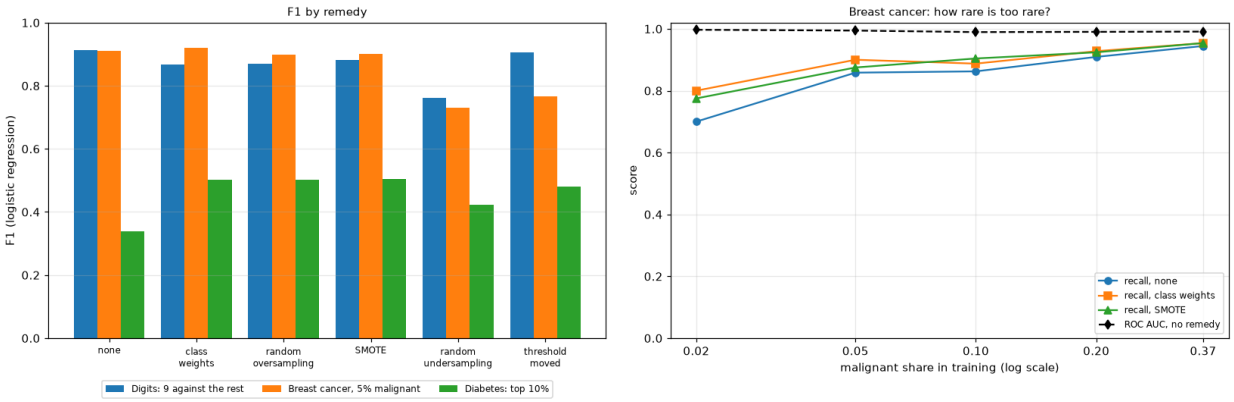
Remedies	None; class weights; random oversampling; SMOTE (k = 5); random undersampling; threshold moved (chosen on 25% of the training rows).
Forest weights	A balanced random forest: each tree's bootstrap draws equally from both classes.
Note	Breast cancer's test part has only about 6 malignant tumours per split, so its recall moves in steps of about 0.17.

ACTUAL OUTPUT

```
Digits: 9 against the rest, logistic:
  remedy                    precision        recall            F1  balanced acc       ROC AUC avg precision
  none                         0.9364        0.8917        0.9128        0.9424        0.9937        0.9659
  class weights                0.8095        0.9398        0.8676        0.9572        0.9922        0.9582
  random oversampling          0.8146        0.9361        0.8694        0.9559        0.9921        0.9590
  SMOTE                        0.8424        0.9324        0.8812        0.9552        0.9913        0.9536
  random undersampling         0.6308        0.9639        0.7609        0.9499        0.9845        0.8916
  threshold moved              0.9243        0.8944        0.9059        0.9428        0.9937        0.9659
Digits: 9 against the rest, forest:
  remedy                    precision        recall            F1  balanced acc       ROC AUC avg precision
  none                         0.9988        0.7417        0.8497        0.8708        0.9896        0.9615
  class weights                0.8202        0.9204        0.8668        0.9488        0.9899        0.9556
  random oversampling          0.9870        0.8065        0.8859        0.9026        0.9946        0.9760
  SMOTE                        0.9891        0.8315        0.9023        0.9152        0.9927        0.9726
  random undersampling         0.7147        0.9481        0.8134        0.9525        0.9888        0.9501
  threshold moved              0.9128        0.8704        0.8869        0.9301        0.9896        0.9615
Breast cancer, 5% malignant, logistic:
  remedy                    precision        recall            F1  balanced acc       ROC AUC avg precision
  none                         0.9829        0.8583        0.9098        0.9287        0.9943        0.9647
  class weights                0.9565        0.9000        0.9212        0.9486        0.9940        0.9611
  random oversampling          0.9423        0.8750        0.8981        0.9356        0.9919        0.9609
  SMOTE                        0.9506        0.8750        0.9019        0.9359        0.9924        0.9622
  random undersampling         0.6404        0.9333        0.7305        0.9449        0.9928        0.9289
  threshold moved              0.9611        0.6917        0.7670        0.8442        0.9943        0.9647
Breast cancer, 5% malignant, forest:
  remedy                    precision        recall            F1  balanced acc       ROC AUC avg precision
  none                         0.9579        0.7250        0.8052        0.8613        0.9725        0.9032
  class weights                0.6384        0.9000        0.7378        0.9339        0.9877        0.9120
  random oversampling          0.9500        0.7833        0.8444        0.8903        0.9828        0.9132
  SMOTE                        0.8874        0.7500        0.7963        0.8720        0.9833        0.9088
  random undersampling         0.5421        0.9333        0.6614        0.9377        0.9894        0.9052
  threshold moved              0.8867        0.6167        0.6840        0.8048        0.9725        0.9032
Diabetes: top 10%, logistic:
  remedy                    precision        recall            F1  balanced acc       ROC AUC avg precision
  none                         0.4759        0.2750        0.3386        0.6193        0.8967        0.4659
  class weights                0.3711        0.7929        0.5023        0.8151        0.8991        0.4784
  random oversampling          0.3677        0.8036        0.5022        0.8185        0.8986        0.4795
  SMOTE                        0.3736        0.7893        0.5056        0.8154        0.8971        0.4876
  random undersampling         0.2990        0.7429        0.4235        0.7676        0.8620        0.4202
  threshold moved              0.4080        0.6607        0.4795        0.7643        0.8967        0.4659
Diabetes: top 10%, forest:
```

ACTUAL OUTPUT

```
  remedy                    precision        recall            F1  balanced acc       ROC AUC avg precision
  none                         0.3841        0.1143        0.1637        0.5446        0.8583        0.4090
  class weights                0.3444        0.7214        0.4647        0.7789        0.8723        0.4271
  random oversampling          0.3934        0.2750        0.3137        0.6119        0.8487        0.3875
  SMOTE                        0.3716        0.4179        0.3878        0.6670        0.8621        0.4254
  random undersampling         0.3057        0.8357        0.4458        0.8043        0.8656        0.4286
  threshold moved              0.3307        0.5679        0.3991        0.7180        0.8583        0.4090
Actual output from this lesson's run.
```



Left: F1 for each remedy with logistic regression. Right: breast cancer recall as the malignant share shrinks, and the ranking quality (AUC).

What it means

There is no single winner. Where the classes separate well (digits and breast cancer with logistic regression), doing nothing is already about as good as anything (0.9128 F1 for digits). Where they overlap (Diabetes), every rebalancing remedy lifts F1 from 0.3386 to about 0.50. SMOTE and class weights are close (0.5056 and 0.5023). Undersampling throws data away and is usually worst. For the forest, weighting matters more: on Diabetes the untreated forest finds 0.1143 of rare cases, the balanced one 0.7214. Moving the threshold helped on Diabetes but hurt on breast cancer, where 25% of a handful of rare rows is too few to set it well.

WHY CHOOSE THIS?

A fair comparison: same splits, same metrics, two kinds of model.

WHY NOT THIS? (TRADEOFFS)

Three small problems; with millions of rows the ranking can differ.

5. Is it just the threshold?

DEFINITION — start here, no prior knowledge assumed

If a remedy only moved the threshold, the ranking of rows (AUC) would not change while recall and F1 would.

REAL-WORLD SCENARIO

Lowering the bar for a smoke alarm makes it ring more, without making it any better at smelling smoke.

ACTUAL OUTPUT

```
ROC AUC and average precision do not depend on the threshold: they measure how well rare rows are
    RANKED above common ones
problem, model                              AUC range across remedies    F1 range
Digits: 9 against the rest, logistic             0.9845 to 0.9937 0.7609 to 0.9128
Digits: 9 against the rest, forest               0.9888 to 0.9946 0.8134 to 0.9023
Breast cancer, 5% malignant, logistic            0.9919 to 0.9943 0.7305 to 0.9212
Breast cancer, 5% malignant, forest              0.9725 to 0.9894 0.6614 to 0.8444
Diabetes: top 10%, logistic                      0.8620 to 0.8991 0.3386 to 0.5056
Diabetes: top 10%, forest                        0.8487 to 0.8723 0.1637 to 0.4647
Actual output from this lesson's run.
```

What it means

Across six remedies, AUC moves by about a hundredth (for example 0.8620 to 0.8991 on Diabetes), while F1 swings widely (0.3386 to 0.5056). The remedies mostly change *where the line is drawn*, not how well the model separates the classes. That is a reason to try the cheapest remedy first: train as usual, then choose a threshold for the trade-off you need.

WHY CHOOSE THIS?

Separates ranking quality from the choice of threshold.

WHY NOT THIS? (TRADEOFFS)

AUC can hide differences that matter at the one threshold you use.

6. What rebalancing does to the probabilities

DEFINITION — start here, no prior knowledge assumed

A model trained as if the rare class were half the data learns that it is half the data. Its probabilities then overstate the rare class.

REAL-WORLD SCENARIO

A doctor trained only on hospital wards overestimates how ill the people in a supermarket are.

ACTUAL OUTPUT

```
mean predicted P(rare) on test rows, against the true rare share; 20 splits
problem                        true share                  none         class weights                 SMOTE
    weights / true
Digits: 9 against the rest         0.1002                0.1008                0.1216               0.1182
    1.21
Breast cancer, 5% malignant        0.0531                0.0473                0.0525               0.0534
    0.99
Diabetes: top 10%                  0.1061                0.1060                0.2670               0.2503
    2.52
rebalancing inflates the predicted rare share most where the classes overlap (Diabetes) and hardly at all
    where they separate cleanly (breast cancer)
Actual output from this lesson's run.
```

What it means

Where the classes overlap, the damage is large: on Diabetes class weights predict a rare share of 0.2670 against a true 0.1061, 2.52 times too many. Where they separate cleanly it is negligible (breast cancer 0.99). Anyone reading these probabilities as risks would need to correct them first.

WHY CHOOSE THIS?

Shows a hidden cost that F1 and AUC do not.

WHY NOT THIS? (TRADEOFFS)

Measured with logistic regression only.

7. How rare is too rare?

DEFINITION — start here, no prior knowledge assumed

Breast cancer again, with the malignant share in training set from 2% to the data's own 37%.

REAL-WORLD SCENARIO

Learning to spot a rare bird from five photographs, or from a hundred and fifty.

ACTUAL OUTPUT

```
 malignant share  rare training rows     none: recall  weights: recall    SMOTE: recall
     none: AUC
            0.02                  5.0           0.7000           0.8000           0.7750
                0.9970
            0.05                 13.0           0.8583           0.9000           0.8750
                0.9943
            0.10                 28.0           0.8625           0.8875           0.9042
                0.9894
            0.20                 62.0           0.9093           0.9278           0.9241
                0.9903
            0.37                148.0           0.9445           0.9539           0.9539
                0.9910
Actual output from this lesson's run.
```

What it means

With 2% malignant there are only 5.0 rare training rows on average. Recall drops to 0.7000, and class weights or SMOTE recover some (0.8000, 0.7750). Yet the ranking stays excellent throughout (0.9970 AUC at 2%): the model can still tell malignant from benign; it is the 0.5 threshold that suits a balanced world.

WHY CHOOSE THIS?

Shows where remedies matter most: very few rare rows.

WHY NOT THIS? (TRADEOFFS)

Five rare rows make every number here noisy.

Interview preparation — technical questions

Q: Why is accuracy misleading on imbalanced data?

A: Always predicting the common class scores its share: 0.8998 on digits, with zero recall.

Q: What does SMOTE do?

A: Creates rare rows by interpolating between a rare row and one of its k nearest rare neighbours.

Q: Class weights or SMOTE?

A: Here they were close (F1 0.5023 and 0.5056 on Diabetes); weights are simpler and add no rows.

Q: Which metrics do you report?

A: Precision, recall, F1 at your threshold, and ROC AUC or average precision for the ranking.

Q: What is the cheapest fix?

A: Keep the model and choose the threshold for the trade-off you need, on held-out rows.

Interview preparation — theoretical questions

Q: Why does rebalancing rarely change AUC?

A: It mostly shifts the scores; the order of rows stays similar (AUC moved by about 0.01 here).

Q: What does rebalancing do to probabilities?

A: Inflates the rare class: 0.2670 predicted against 0.1061 on Diabetes.

Q: When can SMOTE hurt?

A: When rare points are far apart, it fills the space between them with invented rows.

Q: Why must folds be stratified on rare classes?

A: Random folds can have no rare rows at all; Day 118 measures this.

Key takeaway and what's next

> **KEY TAKEAWAY**
>
> On imbalanced data the scoreboard matters more than the remedy. Rebalancing, by SMOTE or by weights, mostly moves the decision threshold: it raises recall where the classes overlap, barely changes how well rows are ranked, and inflates the predicted probabilities. Measure recall, precision and AUC, and try moving the threshold before reaching for synthetic data.

- Diabetes, logistic: recall 0.2750 to about 0.79 with weights or SMOTE; AUC 0.8967 to 0.8991.
- SMOTE and class weights: F1 0.5056 and 0.5023 on Diabetes.
- Rebalanced probabilities: 0.2670 against a true 0.1061 on Diabetes.
- Metrics match scikit-learn to six decimals.

Suggested watch order, if starting from zero

- 1. SMOTE Explained with Example (At A Glance!) — SMOTE on a worked example.
- 2. Handling data imbalance using SMOTE in python (Statistics and Data science) — SMOTE in Python, with random over- and undersampling.

Day 118 preview

- **The Week 17 build: model selection and cross-validation strategies** — k-fold, stratified and time-series splits, using the week's parts.